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Roll No

EX-7202 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Digital Image Processing
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain the significance of sampling and quantization in DIP.
b) Explain about Lossy compression.
2. a) Show that the $N \times N$ cosine transform matrix C is orthogonal. Verify your proof for the case $N = 4$.
b) Prove that an $N \times N$ Haar transform matrix is orthogonal and can be implemented in $O(N)$ operations on an $N \times 1$ vector.
3. Specify the basic components of image processing system. In addition, list steps involved in of image processing system.
4. Differentiate between: Image sensing and acquisition, Image sampling and quantization.
5. a) Show that “For a fixed level of distortion, the number of bits required to transmit the KLT sequence would be less than those required for transmission of the original sequence”.
b) Mention any four properties of the slant transform.
6. a) Write down the basic relationship between pixels.
b) Explain 2D-sampling. What is sampling theorem and aliasing effect?
7. a) Describe the techniques used for color image smoothing.
b) What is the need of gray level slicing in color images?
8. Write short notes on (any two):
 - a) Inverse filtering
 - b) LZW coding
 - c) KL Transform
